

The Number Nobody Was Looking For_Why e Became the Natural Base

Introduction

Opening Question

Begin with the question:

What do astronomy, banking, calculus, radioactive decay, and population growth have in common?

Most people would assume the answer is "nothing."

Yet all of them eventually lead to the same number:

$$e \approx 2.71828$$

Introduce e as a mysterious number that appeared repeatedly in seemingly unrelated areas of mathematics before mathematicians realized they were looking at the same object.

A number discovered by astronomers. Rediscovered by bankers. Named by Euler. And quietly hiding inside calculus.

End with:

This is the story of a number nobody was looking for.

Chapter 1

Before e Existed

Astronomers Were Drowning in Arithmetic

Explain the world before calculators.

Briefly discuss the computational challenges faced in astronomy, navigation, and other scientific fields.

Large calculations were everywhere, and multiplication of large numbers was slow, tedious, and prone to error.

Key Idea

People were not looking for a new branch of mathematics.

They simply wanted faster calculations.

Desired Reader Reaction

"Okay, I understand why someone needed a shortcut."

Creative Elements

- Historical astronomer illustration
 - Multiplication challenge for the reader
-

Chapter 2 John Napier's Shortcut

Introduce John Napier.

Explain that his goal was to simplify difficult calculations, leading to the invention of logarithms.

Turning Multiplication into Addition

Introduce the central idea behind logarithms.

Avoid formal definitions.

Focus on the intuition:

If multiplication can be transformed into addition, calculations become much easier.

Use simple numerical examples.

Creative Element

Show:

$$10 \times 100 = 1000$$

becoming

$$1 + 2 = 3$$

in logarithmic language.

Logarithm Tables

Explain how logarithm tables worked.

Scientists would:

1. Look up logarithms
2. Add values
3. Perform a reverse lookup

Explain that logarithm tables served the role of calculators for centuries.

Mention that students continued using them well into the twentieth century.

Explain that Napier was not searching for e . Although a number closely related to e was already hidden within logarithms, nobody recognized it at the time.

End with suspense.

Several decades later, a banker's question would lead mathematicians back to the same number.

Creative Elements

- Historical logarithm table scan
- Zoomable table example

Fun Fact (Collapsible)

Anti-log

Historically, mathematicians referred to the inverse process as an anti-logarithm.

Today, we simply think of it as exponentiation.

Chapter 3

Bernoulli's Question

Can a Bank Account Grow Forever?

Introduce Jacob Bernoulli.

Briefly mention that his work would eventually lead to the same mysterious number.

Present the problem:

Suppose:

- \$1 initial investment

- 100% annual interest

What happens if interest is compounded more frequently?

Show examples:

Annual:

$$(1 + 1)^1 = 2$$

Quarterly:

$$\left(1 + \frac{1}{4}\right)^4$$

Continue the pattern.

Creative Element

Interactive compound-interest slider.

A Number Appears

Explain that increasing the compounding frequency increases the final amount.

However, each increase becomes smaller than the previous one.

Introduce:

$$\left(1 + \frac{1}{n}\right)^n$$

Explain convergence intuitively.

Reveal:

$$e \approx 2.71828$$

Bernoulli had arrived at the same number through an entirely different problem.

Creative Element

Convergence animation.

Transition

The same number had now appeared in two unrelated contexts.

Yet no one had explicitly identified it.

It remained hidden—until Euler.

Creative Element

Image comparison between the two spirals.

Bernoulli's Obsession with Spirals

Brief sidebar.

Mention the logarithmic spiral.

Tell the famous tombstone story:

Bernoulli requested a logarithmic spiral.

The engraver accidentally carved an Archimedean spiral instead.

Chapter 4

Euler Connects the Dots

Wait... We've Seen This Number Before

Explain that Euler recognized the connection between these apparently unrelated appearances.

This was not a new discovery.

It was a realization.

The number arising from logarithms and the number arising from Bernoulli's limit were actually the same.

Briefly discuss Euler's use of infinite series and Taylor expansions in studying e .

Mention Euler's formula and the famous identity:

$$e^{i\pi} + 1 = 0$$

without turning them into the chapter's main focus.

Creative Element

- Historical timeline showing Napier → Bernoulli → Euler.
- Why Is It Called e ?

Discuss the common myth that Euler named the constant after himself.

Explain that historians generally believe this is unlikely.

Most likely, e was simply the next convenient letter available.

- The Pi-and-e Notation Disaster

Introduce Benjamin Peirce's alternative notation.

Show examples.

Explain why it never gained acceptance.

Keep the tone light and humorous.

- Notation comparison graphic
 - Humorous meme section
-

Transition

Now that the number had a name, mathematicians began uncovering its remarkable properties.

Chapter 5

Why Calculus Loves e

The Only Exponential That Reproduces Itself

Introduce exponential functions:

$$a^x$$

Briefly explain that different bases produce different growth rates.

Since a visual demonstration will be provided, keep the discussion concise.

Show that among all exponential functions, only one satisfies

$$\frac{d}{dx} a^x = a^x$$

namely

$$a = e.$$

This is the real reason e is called the natural base.

Creative Elements

- Base slider
 - Tangent line visualization
 - Derivative comparison graph
-

Why This Matters

Explain that many differential equations describe rates of change.

Because e reproduces itself under differentiation, it naturally becomes the language of continuous growth.

e Is Everywhere

Population Growth

$$P(t) = P_0 e^{rt}$$

Radioactive Decay

$$N(t) = N_0 e^{-\lambda t}$$

Electricity

Brief discussion of RC circuits.

Keep all application sections short.

The focus should remain on the repeated appearance of e , rather than the applications themselves.

Key Message

Whenever a system changes at a rate proportional to its current size, e tends to appear naturally.

The Real Meaning of "Natural"

Ask:

Why isn't base 10 natural?

Why isn't base 2 natural?

Why isn't base 7 natural?

Answer:

Because humans chose those bases.

Nature didn't.

But when mathematicians study continuous growth, differential equations, and rates of change, one base repeatedly appears without anyone choosing it:

$$e$$

That is why it became known as the natural base.

Creative Elements: a fun fact

The 37% Rule

In the optimal stopping problem, the best strategy is to reject approximately

$$1/e$$

of the candidates before making a serious choice.

Keep the tone playful.

You may also include the "rebel" function

$$x^{-1}$$

as a small mathematical Easter egg.

Why Do Different Fields Use Different Logarithms?

Answer a question many students naturally ask.

Explain why logarithm notation varies across disciplines.

Discuss:

- Mathematics
- Engineering

- Computer Science

Explain how:

- ($\ln$)

- ($\log$)

can mean different things depending on context.

Clarify the common conventions used in each field.

Conclusion

The Number That Never Left

Recap the journey:

Astronomy $\rightarrow$ Logarithms $\rightarrow$ Bernoulli $\rightarrow$ Euler $\rightarrow$ Calculus $\rightarrow$ Nature

Emphasize that nobody intentionally searched for e .

It simply kept appearing.

Eventually mathematicians stopped asking:

- | Where did this number come from?

and started asking:

- | Why does it keep appearing everywhere?

End on a reflective note.